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cohort A

//qn. 7.24

```
MariaDB [employeesystem]> create table department(deptNo int(10) not
null, deptName varchar(30), mgrEmpNo int(10), primary key(deptNo));
Query OK, 0 rows affected (0.119 sec)
```

```
MariaDB [employeesystem]> create table employee(empNo int(10) not
Null, primary key(empNo), fName varchar(30), lName varchar(30), DOB
date, sex varchar(2), position varchar(40), deptNo int(10), foreign
key (deptNo) references department(deptNo));
Query OK, 0 rows affected (0.117 sec)
```

```
MariaDB [employeesystem]> alter table department add foreign key
(mgrEmpNo) references employee(empNo);
Query OK, 0 rows affected (0.247 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [employeesystem]> create table project(projNo int(10) not
null, primary key (projNo), projName varchar(40), deptNo int(10),
foreign key (deptNo) references department(deptNo));
Query OK, 0 rows affected (0.118 sec)
```

```
MariaDB [employeesystem]> create table workson(empNo int(10) not
null, projNo int(10) not null, dateWorked date not null, hoursWorked
decimal(10,2), primary key(empNo, projNo, dateWorked), foreign key
(empNo) references employee (empNo), foreign key (projNo) references
project(projNo));
Query OK, 0 rows affected (0.118 sec)
```

```
MariaDB [employeesystem]> alter table employee add constraint
sex_check check(sex in('M','F'));
Query OK, 0 rows affected (0.248 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [employeesystem]> alter table employee add constraint
position_check check(position in('Manager','Team leader','Analyst',
'Software Developer'));
Query OK, 0 rows affected (0.236 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [employeesystem]> alter table workson add constraint
hours_check check(hoursWorked>=0 and hoursWorked<=40);
Query OK, 0 rows affected (0.240 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [employeesystem]> alter table workson modify column
hoursWorked int(3);
Query OK, 0 rows affected (0.232 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [employeesystem]> INSERT INTO department (deptNo, deptName, mgrEmpNo) VALUES
```

```
-> (4, 'Human Resources', NULL),  
-> (17, 'Finance', NULL),  
-> (205, 'Information Technology', NULL),  
-> (412, 'Marketing', NULL),  
-> (999, 'Sales', NULL),  
-> (1200, 'Research', NULL),  
-> (2006, 'Customer Support', NULL),  
-> (3008, 'Operations', NULL),  
-> (4010, 'Procurement', NULL),  
-> (5055, 'Legal', NULL),  
-> (6100, 'Training', NULL),  
-> (7222, 'Quality Assurance', NULL),  
-> (8450, 'Security', NULL),  
-> (9101, 'Administration', NULL),  
-> (12005, 'Business Intelligence', NULL);
```

```
Query OK, 15 rows affected (0.016 sec)
```

```
Records: 15 Duplicates: 0 Warnings: 0
```

```
MariaDB [employeesystem]> INSERT INTO employee
```

```
-> (empNo, fName, lName, DOB, sex, position, deptNo)
```

```
-> VALUES
```

```
-> (101, 'John', 'Mensah', '1980-04-12', 'M', 'Manager', 4),  
-> (207, 'Mary', 'Asante', '1985-08-22', 'F', 'Manager', 17),  
-> (450, 'Daniel', 'Owusu', '1982-02-18', 'M', 'Manager', 205),  
-> (782, 'Linda', 'Boateng', '1988-09-10', 'F', 'Manager', 412),  
-> (1005, 'Michael', 'Agyeman', '1983-11-05', 'M', 'Manager', 999),  
-> (1500, 'Grace', 'Nyarko', '1990-03-15', 'F', 'Team leader', 1200),  
-> (2022, 'Samuel', 'Darko', '1989-12-01', 'M', 'Team leader', 2006),  
-> (3098, 'Patricia', 'Addo', '1991-07-20', 'F', 'Team leader', 3008),  
-> (4011, 'Joseph', 'Ofori', '1987-05-08', 'M', 'Team leader', 4010),  
-> (5100, 'Evelyn', 'Boadu', '1992-01-25', 'F', 'Team leader', 5055),  
-> (6123, 'Richard', 'Kusi', '1994-06-30', 'M', 'Analyst', 6100),  
-> (7555, 'Angela', 'Tetteh', '1993-10-14', 'F', 'Analyst', 7222),  
-> (8800, 'Peter', 'Amankwah', '1995-02-28', 'M', 'Software
```

```
Developer', 8450),
```

```
-> (9301, 'Sandra', 'Frimpong', '1996-04-18', 'F', 'Software
```

```
Developer', 9101),
```

```
-> (15001, 'David', 'Acheampong', '1997-08-09', 'M', 'Software
```

```
Developer', 12005);
```

```
Query OK, 15 rows affected (0.004 sec)
```

```
Records: 15 Duplicates: 0 Warnings: 0
```

```
MariaDB [employeesystem]> INSERT INTO project (projNo, projName, deptNo) VALUES
```

```
-> (11, 'Employee Portal Upgrade', 4),  
-> (145, 'Payroll Automation', 17),  
-> (2088, 'Cloud Migration', 205),  
-> (3001, 'Digital Marketing Campaign', 412),  
-> (4500, 'Sales Analytics Platform', 999),  
-> (5012, 'AI Research Initiative', 1200),  
-> (6200, 'Customer Ticket System', 2006),  
-> (7777, 'Operations Optimization', 3008),
```

```

-> (8900,'Supplier Management System',4010),
-> (9999,'Contract Management Tool',5055),
-> (12010,'E-Learning Platform',6100),
-> (14550,'Product Quality Dashboard',7222),
-> (18000,'Security Monitoring System',8450),
-> (20555,'Office Resource Management',9101),
-> (30000,'Business Intelligence Warehouse',12005);
Query OK, 15 rows affected (0.003 sec)
Records: 15  Duplicates: 0  Warnings: 0

```

```

MariaDB [employeesystem]> UPDATE department
-> SET mgrEmpNo = CASE deptNo
->     WHEN 4 THEN 101
->     WHEN 17 THEN 207
->     WHEN 205 THEN 450
->     WHEN 412 THEN 782
->     WHEN 999 THEN 1005
->     WHEN 1200 THEN 1500
->     WHEN 2006 THEN 2022
->     WHEN 3008 THEN 3098
->     WHEN 4010 THEN 4011
->     WHEN 5055 THEN 5100
->     WHEN 6100 THEN 6123
->     WHEN 7222 THEN 7555
->     WHEN 8450 THEN 8800
->     WHEN 9101 THEN 9301
->     WHEN 12005 THEN 15001
-> END
-> WHERE deptNo IN
->
(4,17,205,412,999,1200,2006,3008,4010,5055,6100,7222,8450,9101,12005
);
Query OK, 15 rows affected (0.015 sec)
Rows matched: 15  Changed: 15  Warnings: 0

```

```

MariaDB [employeesystem]> select * from EmpProject;
Empty set (0.009 sec)

```

```

MariaDB [employeesystem]> INSERT INTO workson
-> (empNo, projNo, dateWorked, hoursWorked)
-> VALUES
-> (101,11,'2025-01-05',8),
-> (207,145,'2025-01-06',7),
-> (450,2088,'2025-01-07',9),
-> (782,3001,'2025-01-08',6),
-> (1005,4500,'2025-01-09',8),
-> (1500,5012,'2025-01-10',10),
-> (2022,6200,'2025-01-11',7),
-> (3098,7777,'2025-01-12',8),
-> (4011,8900,'2025-01-13',9),
-> (5100,9999,'2025-01-14',5),
-> (6123,12010,'2025-01-15',8),
-> (7555,14550,'2025-01-16',7),
-> (8800,18000,'2025-01-17',10),

```

```

-> (9301,20555,'2025-01-18',6),
-> (15001,30000,'2025-01-19',8),
->
-> (450,30000,'2025-02-01',12),
-> (8800,2088,'2025-02-02',15),
-> (9301,7777,'2025-02-03',11),
-> (1500,14550,'2025-02-04',14),
-> (6123,4500,'2025-02-05',13),
-> (3098,20555,'2025-02-06',10),
-> (15001,5012,'2025-02-07',16),
-> (2022,18000,'2025-02-08',9),
-> (4011,12010,'2025-02-09',12),
-> (7555,3001,'2025-02-10',8),
->
-> (101,11,'2025-03-01',7),
-> (101,11,'2025-03-02',8),
-> (101,11,'2025-03-03',6),
-> (207,145,'2025-03-01',9),
-> (207,145,'2025-03-02',8);

```

Query OK, 30 rows affected (0.003 sec)
Records: 30 Duplicates: 0 Warnings: 0

MariaDB [employeesystem]> select * from EmpProject;
Empty set (0.016 sec)

empNo	projNo	totalHours
101	11	29
207	145	24
450	2088	9
450	30000	12
782	3001	6
1005	4500	8
1500	5012	10
1500	14550	14
2022	6200	7
2022	18000	9
3098	7777	8
3098	20555	10
4011	8900	9
4011	12010	12
5100	9999	5
6123	4500	13
6123	12010	8
7555	3001	8
7555	14550	7
8800	2088	15
8800	18000	10
9301	7777	11
9301	20555	6
15001	5012	16
15001	30000	8

//qn7.25

```
MariaDB [employeesystem]> create view FemaleLedProjects as select
project.* from project,employee,department where
department.mgrEmpNo=employee.empNo and employee.sex='F' and
department.deptNo=project.deptNo order by projNo;
Query OK, 0 rows affected (0.108 sec)
```

//qn 7.26

```
MariaDB [employeesystem]> create view EmpWork as select
workson.empNo, fName, lName, projName, hoursWorked from employee,
project, workson where employee.empNo=workson.empNo and
project.projNo=workson.projNo;
Query OK, 0 rows affected (0.090 sec)
```

//qn 7.27

```
MariaDB [employeesystem]> create view EmpProject(empNo, projNo,
totalHours) AS select w.empNo, w.projNo, SUM(hoursWorked) FROM
Employee e, Project p, WorksOn w where e.empNo = w.empNo and
p.projNo = w.projNo GROUP BY w.empNo, w.projNo;
Query OK, 0 rows affected (0.100 sec)
```

```
MariaDB [employeesystem]> select projNo from EmpProject where
projNo='SCCS';
Empty set (0.006 sec)
```

```
MariaDB [employeesystem]> select count(projNo) from EmpProject where
empNo='E1';
```

count(projNo)
0

1 row in set (0.001 sec)

```
MariaDB [employeesystem]> select empNo, totalHours from EmpProject
group by empNo;
```

empNo	totalHours
101	29
207	24
450	9
782	6
1005	8
1500	10
2022	7
3098	8
4011	9
5100	5
6123	13
7555	8
8800	15
9301	11
15001	16

```
+-----+-----+
15 rows in set (0.001 sec)
```

//qn 7.28

As a materialized view i would use a table to store the information and periodically add to it the new values stored on the base table. If the view did not have a Distinct constraint, apart from that it meets every other condition needed for it to be maintained without having access to the underlying base table. Each roww innput must also be able to be traced back to the base table